EART11300 Practical Class 2 or 3

At the end of this class you should be able to:

- Describe the parts of a published geological map
- Describe what a contour is and use topographic to find different features on a map
- Construct a topographic profile
- Open Digimap and use it to find Grid References

Maps

A map that shows the physical features of the surface of the Earth, including the heights of the physical features above a specified datum level is called a topographic map.

Topographic maps show the heights of features, or relief, using contour lines. A contour line is a line that links heights of equal elevation and is always horizontal.

A geological map shows the lithostratigraphic formations that would occur at the ground surface if there were no superficial coverings such as soil or tarmac or buildings. Sometimes we informally call these formations geological units. The geology is superimposed on a topographic map which will show the contours and other physical features which will allow us to interpret the geology.

In the UK geological maps are produced and published by the British Geological Survey

http://www.bgs.ac.uk/

And the topographic maps are produced and published by the Ordnance Survey

https://www.ordnancesurvey.co.uk/

You will see as we progress through the course that the majority of the rocks that occur on the maps are sedimentary. That means we can use the Laws of Superposition and Original Horizontality to start to interpret them.

A bed is the smallest stratigraphical division that we use and it refers to individual layers that were deposited from a flow event and often over a short period of time. The bedding surface will often have formed the sediment surface at some point in the past. You will often hear us talk about bedding planes – the planes are the

boundaries between the beds and we use the orientation of the planes to work out the geological structures and geological history of an area.

A formation is a collection of beds that share some similarities e.g. the rock type, or sedimentary structures that show they were deposited in a particular environment. A formation must be of sufficient size that it is visible on the map and distinct from other formations but the contact does not have to be sharp or distinct itself e.g. a formation could change when the proportion of beds of conglomerate exceed 50%.

On a geological map the line that is drawn between 2 different formations is called the contact. We can show how confident we are about the position of a contact by using different symbols. A solid line means that we are absolutely confident in the position of the line. A dashed line means that there is some uncertainty and if we are really unsure then we can use a dashed line with question marks.

A solid line – absolutely confident

A dashed line – some uncertainty

Outcrop and exposure

Exposure is where the rock is exposed on the surface and you could stand on it and measure the orientation of the bedding planes.

Outcrop is what we see on a geological map and it shows where a formation would occur if it were exposed at the surface.

Looking at outcrop patterns on geological maps

The outcrop pattern of a geological formation shows the intersection of the 3-dimensional shape of the rock with the land surface.

So, for example, if the beds within a formation are horizontal then the boundaries between formations will be parallel to the topographic contours.

A vertical feature, like a fault or an igneous dyke will outcrop as a straight line crossing any topographic contours.

If the beds are dipping then the contacts will wave across the land surface cutting across the topographic contours in a way we will be able to work out.

Strike and Dip

Strike is the azimuth (compass bearing) of a horizontal line on a bed. This line will also be perpendicular to the direction of the steepest angle of dip. This is always given as a 3-figure number e.g. 180 for a bed striking North/South

Dip is the angle from the horizontal of the steepest gradient on the bedding surface. This is always given as a 2-figure number e.g. a vertical bed is 90° and a horizontal bed has a dip of 00°. You must also give the direction that the bed is dipping as a compass direction e.g. East.

The full reading takes the form of 180/24E for example.

Strike and Dips are marked on maps using a combination of symbols and numbers.

Strike and dip are measured using a compass clinometer or nowadays there are various phone apps but you must be able to use a compass clinometer first.

Working out Dip direction from outcrop patterns

We can work out the direction of dip if we know the relative ages of the rocks. To do this we use the stratigraphic column found with all geological maps. The strat column shows the orders of the formations visible on the map with the oldest at the bottom of the column and the youngest at the top. If the beds are dipping then the dip direction goes from older to younger (this doesn't work if the beds are upside down but we will start with the simple cases first)

Stratum Contours

Stratum contours are contours that can be drawn on a geological surface, this is normally a bedding surface but it could be a fault plane surface or the surface of a coal seam. Like topographic contours they are drawn at specific heights above a datum level. They allow us to represent the 3 dimensional shape of the rock surfaces below ground or even above ground before they were eroded to the present land surface.

If the geological surface is planar then the stratum contours will be straight, parallel and equidistant.

The strike of the surface is parallel to the strike of the stratum contours.

If the stratum contours are curved then the strike of the geological surface will vary and follow the curves.

If the separation between the stratum contours varies then the dip of the geological surface varies.

Activities:

1. Look at the topographic map and identify a valley, a steep slope, a gentle slope, a ridge, a valley and a hill. Draw examples of these in your lab book.

2. Identify a place on the map that shows a valley, a ridge or a hill and construct a topographic profile on the graph paper provided. Your profile should be between 15-20cm in length. You must record the grid references at both ends of your profile. Your profile (or cross section) must be the same vertical scale as the horizontal scale of the map.

3. Draw out the points of a compass with the bearings for north, south, east and west in degrees in your lab book.

4. Look at the geological map Sheet 318/333 Brighton & Worthing and answer the questions on the separate question sheet.

5. Open Digimap on you laptop or phone or borrow one of our iPad. Follow the instructions on the BB page to find the British National Grid values for the Eastings and Northings of where your home is e.g 385708, 378225. Go to the teaching room pc and enter these values in the Excel file that will be open. We will use the class values to ask Digimap to plot on a map of the UK where our homes are across the country. If you are not from the UK then you can enter the values for either Alderley Edge or Mam Tor.